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Quantification of electromyographic activity during REM sleep in Chinese narcolepsy patients and healthy controls: a comparative study

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Abstract

Study objectives This study aimed to assess and quantify rapid eye movement (REM) sleep electromyographic (EMG) activity in narcolepsy patients.

Methods The study involved 95 participants, including 25 with narcolepsy type 1 (NT1), 36 with narcolepsy type 2 (NT2), and 34 healthy controls (HC). We visually analyzed phasic, tonic, and "any" muscle activity for the submentalis (SM) and phasic for the anterior tibialis (AT) by both 3-s REM mini-epochs (3sRME) and 30-s REM epochs (30sRE) methods. Group metrics of EMG activity indices in 3sRME and positive REM sleep without atonia (RWA) indices in 30sRE (AASM rules) were analyzed. Receiver operating characteristic (ROC) curves determined the best cutoff thresholds for narcolepsy.

Results All median SM and AT EMG activity indices in 3sRME were significantly higher in NT1 and NT2 compared to HC ($P<0.05$). In the 30sRE method, positive RWA with phasic SM, total SM, and combined SM + AT indices were significantly higher only in NT1 versus HC in all analyses. The 3sRME phasic AT index had the highest area under the curve (AUC) of 0.884 (cutoff > 1.74%) for narcolepsy, 0.920 (cutoff > 2.03%) for NT1, and 0.859 (cutoff > 1.74%) for NT2. In the 30sRE method, the positive RWA with combined SM + AT index had the highest AUC of 0.691 (cutoff > 4.37%) for narcolepsy.

Conclusions The study demonstrated increased levels of REM sleep EMG activity in narcoleptic patients compared to HC, with more pronounced abnormalities in NT1 than NT2, suggesting its potential as an alternative biomarker for narcolepsy.

Keywords Narcolepsy · Polysomnography · REM sleep without atonia · Electromyography · Diagnostic biomarkers

Abbreviations

3sRME	3–second REM mini–epochs
30sRE	30–second REM epochs
AASM	American Academy of Sleep Medicine
AHI	Apnea–Hypopnea Index
AOR	Adjusted Odds Ratio
AT	Anterior tibialis (muscle)
AUC	Area Under the Curve
BMI	Body Mass Index
CCSD	Chinese Clinical Sleep Database

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CI	Confidence Interval
CSF	Cerebrospinal Fluid
EDS	Excessive Daytime Seepiness
EMG	Electromyography
HC	Healthy Controls
ICSD	3–TR–International Classification of Sleep Disorders–Third Edition, Text Revision
IQR	Interquartile Range
KW	Kruskal–Wallis Test
MSLT	Multiple Sleep Latency Test
NT1	Narcolepsy Type 1
NT2	Narcolepsy Type 2
NREM	Non–Rapid Eye Movement (sleep)
PLMs	Periodic Limb Movements
PSG	Polysomnography
REM	Rapid Eye Movement (sleep)
REML	REM Sleep Latency
RBD	REM Sleep Behavior Disorder
ROC	Receiver Operating Characteristic
RWA	REM Sleep Without Atonia
SE	Sleep Efficiency
SINBAR	Sleep Innsbruck Barcelona (scoring method)
SL	Sleep Latency
SM	Submentalis (muscle)
SOREMP	Sleep Onset REM Period
TST	Total Seep Time
WASO	Wake After Sleep Onset

Introduction

Narcolepsy is a chronic neurological disorder characterized by excessive daytime sleepiness (EDS) and dissociated rapid eye movement (REM) sleep phenomena, such as cataplexy, hypnagogic/hypnopompic hallucinations, and sleep paralysis [1, 2]. Although REM sleep is characterized by muscle atonia, individuals with narcolepsy often display excessive twitching and persistent muscle tone, underscoring the disorder's unique neurophysiological profile [3–6]. Quantifying these abnormalities using electromyographic (EMG) activity may provide an alternative marker for narcolepsy, offering potential diagnostic utility [3, 5].

Previous studies have employed varying methodologies to quantify REM sleep EMG activity in narcolepsy, with some focusing on chin muscles [3, 6] and others including both chin and limb muscles [5]. In addition, some studies have used 20-s epochs for tonic activity and 2-s epochs for phasic activity [6], while others have utilized 30-s epochs for tonic and 3-s epochs for phasic activity [3, 5]. Proposed cutoffs for abnormal muscle activity in narcolepsy, such as ≥1.48% for tonic and ≥2.0% for phasic chin muscle activity [6] or ≥5.97% for combined chin EMG activity and ≥8.37% for combined chin and limb activity [5], highlight significant variability across studies. These methodological differences, along with inconsistent cutoffs, underscore the need for a more unified approach to quantify REM sleep EMG activity in narcolepsy.

Furthermore, there is a lack of comprehensive data comparing narcolepsy type 1 (NT1), type 2 (NT2), and healthy controls (HC). Specifically, studies using both the percentage of total 3-s REM mini-epochs (3sRME) with EMG activity (Sleep Innsbruck Barcelona [SINBAR] scoring method) [7] and the percentage of 30-s REM epochs (30sRE) positive for REM sleep without atonia (RWA) under American Academy of Sleep Medicine (AASM) criteria [8] are lacking. These indices for chin (submentalis, SM), limb (anterior tibialis, AT), and a combined SM+AT measure would provide valuable markers using routine polysomnography (PSG) data.

This study quantified EMG activity during REM sleep in Chinese narcolepsy patients and compared indices in NT1 and NT2 with HC.

Methods

The methods used for the case-control study are described below.

Study design and population

A retrospective case-control study examined muscle activity during REM sleep in narcolepsy (NT1 and NT2) patients and HC (Fig. 1). The study conforms to the principles outlined by the Declaration of Helsinki and utilized data acquired through the Chinese Clinical Sleep Database (CCSD) [9]. The CCSD has been approved by the ethical committee of Nanfang Hospital, Southern Medical University.

Participant selection

Data for narcolepsy cases ($n=82$) were identified from the CCSD. These patients were diagnosed with NT1 or NT2 based on the International Classification of Sleep Disorders, Third Edition (ICSD-3) [10], between January 2018 and December 2023. Patients diagnosed with comorbid REM Sleep Behavior Disorder (RBD) according to ICSD-3 criteria were excluded. The presence of cataplexy was confirmed by a sleep medicine specialist, and all NT1 patients had a documented history of cataplexy. Exclusion criteria included an apnea-hypopnea index (AHI) > 15 events/hour and poor EMG data quality. A total of 21 cases were excluded: 14 due to AHI > 15 and 7 due to poor EMG data quality. Among NT1 patients, 44% (11/25) were medication-free, while

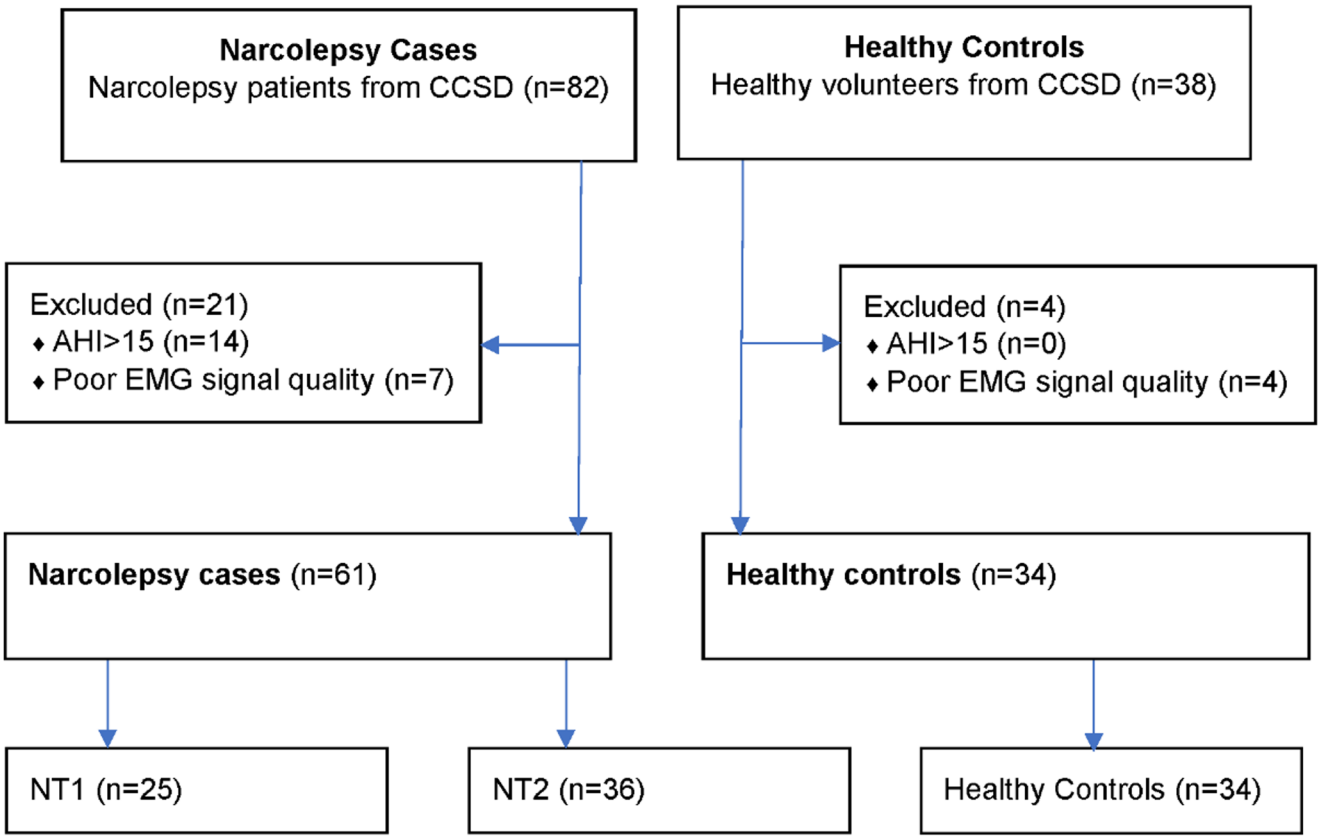

Fig. 1 Study Flow Diagram. CCSD: Chinese Clinical Sleep Database; AHI: Apnea-Hypopnea Index; NT: Narcolepsy Type; EMG: Electromyography

16% (4/25) used antidepressants, 20% (5/25) used stimulants, and 20% (5/25) used both antidepressants and stimulants. For NT2, 61.1% (22/36) were medication-free, with 30.6% (11/36) on stimulants and 8.3% (3/36) on both antidepressants and stimulants.

HC group members ($n=38$) were healthy volunteers selected from CCSD. Inclusion criteria were: medication-free individuals with no clinical diagnosis of mental disorder and no history suggestive of narcolepsy or other sleep disorders. Exclusion criteria included an AHI > 15 and poor EMG data quality. Four HCs were excluded due to poor EMG data quality. The final dataset for analysis comprised 25 NT1 patients, 36 NT2 patients, and 34 HC.

Nocturnal PSG and MSLT

Nocturnal PSG assessments were conducted at the participating medical centers in CCSD using digital systems. Standard PSG measurements included electrooculography (EOG), electroencephalography (EEG), surface EMG from the SM and AT muscles, electrocardiography (ECG), assessments of nasal and oral airflow, thoracic and abdominal respiratory effort, oxygen saturation, and synchronized audio-video recording. The multiple sleep latency test (MSLT) was conducted in all narcolepsy patients, consisting of five scheduled naps at two-hour intervals starting at 9:00 AM, following an overnight PSG. Healthy controls (HC) underwent nocturnal PSG only; the MSLT was not performed in this group. Sleep stages and related events were manually scored by certified technicians according to the American Academy of Sleep Medicine (AASM) guidelines [8].

Muscles investigated

The quantification of surface muscle activity was done in two key muscle groups: the SM and the AT. EMG recordings were stored, viewed, and scored using the Compumedics Profusion Sleep Software, version 4.50.

Muscle activity analysis

The criteria used for muscle activity quantification were based on recommendations by AASM and the SINBAR Group [7, 8]. After identifying REM epochs, we carefully excluded artifacts and EMG tone increases caused by respiratory events to ensure accurate quantification of REM-related muscle activity [7, 11, 12]. For submentalis (SM) muscle activity, we scored all phasic and tonic EMG activity except when associated with respiratory-related arousals,

gross body movements, electrical or mechanical artifacts, or bruxism. For anterior tibialis (AT) muscle activity, we scored all phasic EMG activity except when occurring within periodic limb movement (PLM) series or associated with respiratory events, gross body movements, or artifacts.

3sRME analysis

Each 30sRE was divided into ten 3sRME, where phasic and "any" SM muscle activity were evaluated. The REM period boundaries for SINBAR analysis were determined by the onset of Stage R and termination of Stage R in 3sRME. After this phasic, or "any" EMG activity was scored within each 3sRME. The phasic activity was defined as transient muscle bursts (0.1–5.0 s). "Any" activity includes all detectable muscle movements regardless of duration. To score phasic and "any" muscle activity amplitude of at least twice the background activity was required. Phasic bursts were scored when both types of activity occurred within the same 3sRME. The end of the phasic burst was defined as a return to the baseline or an interburst interval of >250 ms. The percentage of 3sRME with phasic AT activity and combined "any" SM+AT activity was also recorded. These muscle activities in 3sRME were defined as EMG activity. EMG activity index for 3sRME was calculated as follows:

$$REM\ Sleep\ EMG\ activity\ Index = \frac{Total\ number\ of\ 3sRME\ with\ EMG\ activity}{Total\ number\ of\ 3sRME\ analysed}\ x\ 100\%$$

The EMG activity index was calculated separately for phasic SM, "any" SM, and phasic AT in 3sRME, with an additional combined index for "any" SM or phasic AT (SM+AT).

30sRE analysis

For the 30sRE, epochs positive for RWA were analyzed using AASM criteria [8]. "RWA SM Tonic" was scored when sustained EMG activity (≥15 s duration, amplitude>2 times background or >10μV) was present. Background activity was defined based on the current stage R atonia or the lowest amplitude in adjacent NREM. "RWA SM Phasic" was scored if five or more 3sRME had phasic SM activity in 30sRE. "RWA SM" was classified when 30sRE showed either "RWA SM Tonic", "RWA SM Phasic", or ≥5 3sRME with "any" SM activity. "RWA AT" was scored if≥5 3sRME of phasic AT activity was present in either limb in 30sRE. These positive epochs in 30sRE were defined as RWA. RWA index for 30sRE was calculated as follows:

$$RWA\ Index = \frac{Total\ number\ of\ RWA\ positive\ 30sRE}{Total\ number\ of\ 30sRE\ \ analysed}\ x\ 100\%$$

The RWA index was calculated separately for "RWA SM tonic", "RWA SM phasic", "RWA SM" (tonic, phasic, or "any"), and "RWA AT", with an additional combined index for "RWA SM+AT".

Statistical methods

Descriptive statistics are presented as means (standard deviation) for parametric continuous variables or medians (interquartile range [IQR]) for non-parametric continuous variables, and frequencies (percentages) for categorical variables. Normality was assessed using the Shapiro-Wilk test and visual inspection of Q-Q plots. Comparisons between NT1, NT2, and HC were made using parametric tests (ANOVA with Bonferroni post-hoc tests) for continuous variables with normal distributions, and non-parametric tests (Kruskal-Wallis H tests with Dunn post-hoc tests) for continuous variables that were not normally distributed. Chi-square or Fisher-Freeman-Halton tests were used for categorical variables. Multiple logistic regression adjusted for age, gender, body mass index (BMI), periodic leg movement index (PLMI), and AHI evaluated EMG activity and RWA indices. Receiver operating characteristic (ROC) curves assessed the performance of EMG and RWA in differentiating narcolepsy from controls. A significance level of 0.05 was used for all analyses. Significance values adjusted by the Bonferroni correction for multiple tests were reported for post-hoc tests. Data were analyzed using IBM SPSS Statistics for Windows, Version 27.0. Armonk, NY: IBM Corp and MedCalc® Statistical Software version 22.023 (MedCalc Software Ltd, Ostend, Belgium; https://www.medcalc.org; 2024).

Results

Characteristics of cases and controls

REM sleep EMG data from 95 participants (25 NT1, 36 NT2, and 34 HC) were analyzed. Table 1 presents the demographic, sleep characteristics, REM sleep EMG activity, and RWA indices of the sample (Table 1).

Demographic

The median age of participants was 18 years (IQR: 15–27), with no significant differences across groups ($p=0.078$). Female distribution was comparable (46.3%, $p=0.768$). However, NT2 participants had a significantly higher BMI than HC ($p=0.019$).

Table 1 Demographic, sleep characteristics and REM sleep muscle activity indices in narcolepsy and HC

Characteristic	NT1 (n=25)	NT2 (n=36)	HC (n=34)	Statistics
Age (years)	18(15, 28.50)	17 (15, 19)	26 (14.75, 29.25)	KW=5.11, p=0.078
Female, n (%)	12 (48)	15 (41.7)	17 (50)	χ^2=0.53, p=0.768
BMI (kg/m²)	23.32±3.30	24.32±2.92[c]	21.93±4.36[c]	F=3.89, p=0.024
PSG				
TST (min)	442.50 (391, 475.50)	454.25(415.13,483.75)	445(417.88,472.25)	KW=1.34, p=0.512
SL (min)	7.50 (2.25, 15)	7 (2.13, 17)	11 (4.38, 21.88)	KW=2.31, p=0.316
WASO (min)	38.50 (21.75, 73.25)	28.50 (13.25, 51.75)	30.75 (19.38, 45.13)	KW=1.92, p=0.384
SE (%)	90.30 (84.20, 94.50)	92.70 (87, 94.93)	91.10 (87.65, 94.10)	KW=1.41, p=0.494
REML (min)	61 (2.75, 142.75)	58(1.63, 106.75) [c]	84 (66.13, 137.75) [c]	KW=9.47, p=0.009
REM length (min)	82.30±20.54	95.03±34.15	97.99±22.19	F=2.64, p=0.077
nSOREMP, n (%)	8 (32) [b]	9 (25) [c]	0 (0)[b, c]	χ^2=14.65, p<0.001
AHI (events/hr)	1.20 (0.60, 3.80)	1 (0.35, 3.22)	1 (0.25, 2.53)	KW=1.69, p=0.429
PLMI (events/hr)	3.90 (0.85, 12.40) [b]	1.40 (0, 3.68)	0 (0, 2.70) [b]	KW=12.63, p=0.002
Stage N1 (% TST)	15.70 (7.80, 27.70) [a]	7.70 (4.53, 15.13) [a]	9 (7.33, 12.48)	KW=9.38, p=0.009
Stage N2 (% TST)	47.35±10.73	50.69±10.05	50.05±8.58	F=0.93, p=0.399
Stage N3 (% TST)	15.09±10.75	18.03±7.93	17.87±7.74	F=0.99, p=0.372
Stage R (% TST)	19.30±4.73	20.90±6.56	22.04±3.99	F=1.94, p=0.149
MSLT				
SOREMP (% naps)	80 (60, 100)	80 (60, 100)	-	MWU=447.5, p=0.970
Mean SL (min)	3.10 (1.15, 4.80)	3.75 (1.44, 6.68)	-	MWU=526, p=0.265
3sRME				
3sRME, n	1640.56±411.67	1895.89±682.53	1955.94±443.17	F=2.66, p=0.075
Phasic SM %	8.51 (4.38, 11.23) [b]	5.24 (3.61, 7.95) [c]	3.03 (1.74, 5.11) [b, c]	KW=21.76, p<0.001
"Any" SM %	10.60 (4.61, 13.94) [b]	6.09 (3.96, 8.61) [c]	3.58 (1.91, 6.15) [b, c]	KW=20.91, p<0.001
Phasic AT %	3.95 (2.36, 8.47) [b]	2.56 (1.73, 4.85) [c]	0.95 (0.4, 1.53) [b, c]	KW=40.09, p<0.001
"Any" SM+AT %	12.56 (8.68, 19.75) [b]	9.00 (6.08, 13.68) [c]	4.02 (2.71, 7.08) [b, c]	KW=29.59, p<0.001
30sRE				
30sRE, n	164.60±41.09	190.06±68.30	195.97±44.39	F=2.64, p=0.077
RWA SM Tonic %	0 (0, 1.60)	0 (0, 0.66)	0 (0, 0)	KW=5.97, p=0.051
RWA SM Phasic %	1.91 (0.23, 5.23) [b]	0.99 (0, 2.60)	0.2 (0, 1.6) [b]	KW=8.57, p=0.014
RWA SM %	5.04 (0.84, 8.41) [b]	1.41 (0.09, 4.67)	0.54 (0, 2.15) [b]	KW=11.59, p=0.003
RWA AT %	0 (0, 1.81) [b]	0 (0, 0.45)	0 (0, 0) [b]	KW=10.63, p=0.005
RWA SM+AT %	6.03 (1.5, 9.06) [b]	1.64 (0, 6.13)	0.54 (0, 2.37) [b]	KW=13.8, p=0.001

Note:Values are presented as median (interquartile range), n (%), or mean±standard deviation

NT Narcolepsy type, *HC* Healthy controls, *BMI* Body mass index, *PSG* Polysomnography, *TST* Total sleep time, *SL* Sleep latency, *SE* Sleep efficiency, *WASO* Wake time after sleep onset, REM Rapid eye movement, *REML* REM sleep latency *AHI* Apnea-hypopnea index, *PLMI* Periodic limb movement index, *SOREMP* Sleep onset REM period, *nSOREMP* SOREMP in nocturnal PSG, *min* Minutes, *SM* Submentalis muscle, *AT* Anterior tibialis muscle, *RWA* REM sleep without atonia, *3sRME* 3-second REM mini-epoch, *30sRE* 30-second REM epoch

F ANOVA, *KW* Kruskal Wallis Test, χ^2 Chi-square test, *MWU* Mann–Whitney U test

[a, b, c] Groups with different superscripted letters are significantly different

Sleep continuity and architecture

Total sleep time (TST), sleep latency (SL), wake time after sleep onset (WASO), and sleep efficiency (SE) did not differ significantly between groups. REM sleep latency (REML) was shorter in NT2 compared to HC (p=0.008), and sleep onset REM periods (SOREMPs) were significantly more frequent in NT1 and NT2 than in HC (p<0.001).

NT1 participants spent a higher percentage of TST in stage N1 compared to NT2 (p=0.009). Stages N2, N3, and REM sleep percentages showed no significant differences across groups (Table 1). The PLMI was significantly higher in NT1 than in controls, with no significant differences between NT1 and NT2 or between NT2 and controls.

EMG activity and RWA quantification

Patients with narcolepsy (NT1 and NT2) showed significantly higher rates of phasic SM, "any" SM, phasic AT, and combined "any" SM+AT activity in 3sRME indices compared to HC (All p<0.05) (Table 1). No significant differences were found between NT1 and NT2 in these measures (p>0.05).

In the 30sRE, NT1 exhibited higher "RWA SM phasic" ($p=0.012$), "RWA SM" ($p=0.002$), "RWA AT" ($p=0.005$), and combined "RWA SM+AT" ($p=0.001$) than HC. NT2 did not differ significantly from HC or NT1 in any 30sRE indices.

In multiple logistic regression, after adjusting for age, gender, BMI, PLMI, and AHI, differences in EMG activity indices in 3sRME between narcolepsy/NT1/NT2 and HC were significant. However, in 30sRE, only the total narcolepsy group and NT1 showed significant differences in "RWA SM phasic", "RWA SM", and "RWA SM+AT" indices compared to HC (Fig. 2).

ROC curve analysis of EMG activity and RWA

Table 2 highlights the discriminative performance, cut-off values, sensitivity, specificity, and Youden index of phasic and "any" EMG activity, as well as RWA indices across four models: Model 1 (Narcolepsy vs. HC), Model 2 (NT1 vs. NT2), Model 3 (NT1 vs. HC), and Model 4 (NT2 vs. HC).

In Model 1 (Narcolepsy vs. HC), the 3sRME Phasic AT emerged as the strongest differentiator (AUC=0.884, 95% CI [0.802, 0.941]), followed by the "any" SM+AT (AUC=0.820, 95% CI [0.728, 0.891]). In contrast, the 30sRE RWA indices showed lower discriminatory power, with all AUC values below 0.7.

For Model 2 (NT1 vs. NT2), the 3sRME "any" SM+AT demonstrated the highest discriminatory performance (AUC=0.660, 95% CI [0.527, 0.776]), while other indices showed relatively lower AUC values, with several remaining non-significant.

In Model 3 (NT1 vs. HC), the 3sRME Phasic AT achieved the highest AUC (AUC=0.920, 95% CI [0.819, 0.975]), showing excellent sensitivity and specificity. Among the 30sRE indices, "RWA SM+AT" had the best performance (AUC=0.781, 95% CI [0.654, 0.878]).

For Model 4 (NT2 vs. HC), the 3sRME Phasic AT again showed the strongest performance (AUC=0.859, 95% CI [0.756, 0.931]), with other 3sRME indices also performing well. Despite the lack of significance in group comparisons and logistic regression, ROC curve analysis revealed that "RWA SM Phasic" had significant AUC values for differentiating NT2 from HC ($p<0.05$). However, most 30sRE indices did not significantly distinguish NT2 from HC, as indicated by non-significant p-values or not yielding meaningful cutoff values.

Figure 3 visually highlights the ROC curves of the best-performing indices across all comparisons.

Discussion

Our study demonstrates that EMG activity during REM sleep, measured using 3sRME was significantly elevated in NT1 and NT2 compared to HC, including phasic and "any" SM activity, phasic AT activity, and their combined measures. In our study, phasic AT activity showed the strongest ability to differentiate narcolepsy from HC, outperforming RWA indices quantified using 30sRE. These findings underscore the importance of methodological approaches in quantifying REM sleep muscle activity in narcolepsy, as they can lead to differing objective endpoints.

The elevated REM muscle activity in narcolepsy is tied to its underlying neuropathology [1, 2, 4, 13]. NT1 is strongly associated with the loss of hypocretin-producing neurons in the hypothalamus [14, 15], which disrupts the brain's ability to suppress muscle activity during REM sleep [3, 16]. This hypocretin deficiency leads to dysregulated transitions between sleep-wake states [17], allowing unwanted muscle

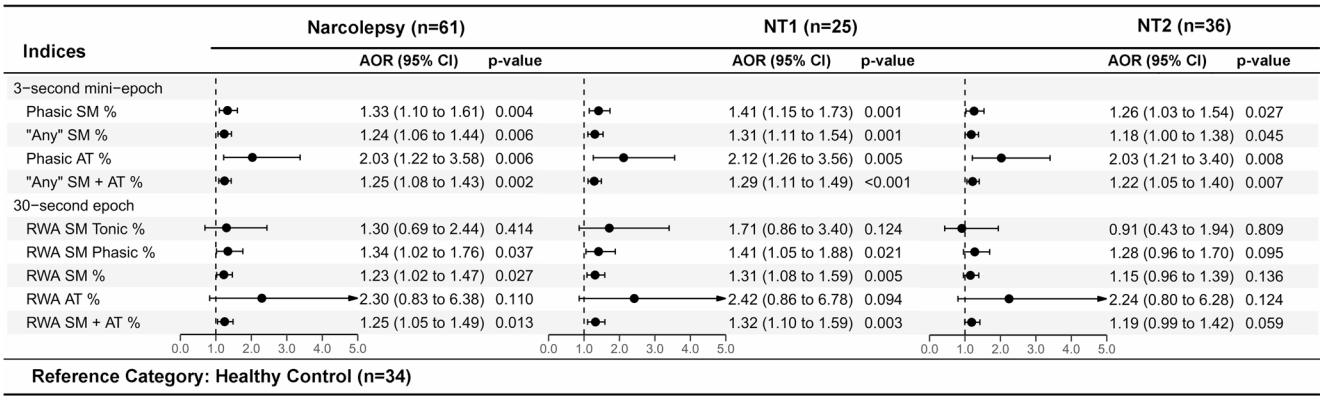

Fig. 2 Regression analysis of REM sleep muscle activity indices in narcolepsy compared to healthy controls. NT: Narcolepsy type; SM: Submentalis muscle; AT: Anterior tibialis muscle; REM: Rapid Eye Movements; RWA: REM sleep without atonia; AOR: Adjusted Odds Ratio; CI: Confidence interval AOR values were adjusted for age, gender, body mass index, apnea-hypopnea index and periodic limb movement index in multiple logistic regression for each index. The narcolepsy (NT1+NT2) group was compared with healthy controls using binary logistic regression. NT1 and NT2 were separately compared with healthy controls using multinomial regression

Table 2 ROC curve analysis of REM sleep muscle activity indices across groups

Characteristics	Cutoff	Sensitivity	Specificity	Youden index J	AUC (95% CI)
Narcolepsy (NT1 + NT2) versus HC					
3-second mini-epoch					
Phasic SM%	> 2.65	93.44	47.06	0.405	0.771 (0.673 to 0.851)
"Any" SM%	> 6.84	55.74	85.29	0.410	0.763 (0.664 to 0.844)
Phasic AT%	> 1.74	78.69	88.24	0.669	0.884 (0.802 to 0.941)
"Any" SM + AT%	> 7.46	70.49	82.35	0.528	0.820 (0.728 to 0.891)
RWA SM Tonic%	> 0.00	36.07	85.29	0.214	0.607 (0.501 to 0.705)
RWA SM Phasic%	> 0.61	62.30	67.65	0.299	0.662 (0.558 to 0.756)
RWA SM%	> 2.16	52.46	79.41	0.319	0.676 (0.573 to 0.769)
RWA AT %	> 0.00	34.43	94.12	0.285	0.644 (0.539 to 0.739)
RWA SM + AT%	> 4.37	44.26	91.18	0.354	0.691 (0.588 to 0.782)
NT1 versus NT2					
3-second mini-epoch					
Phasic SM%	> 7.72	60.00	75.00	0.350	0.650 (0.517 to 0.768)
"Any" SM%	> 8.48	68.00	75.00	0.430	0.659 (0.526 to 0.775)
Phasic AT%	> 4.96	48.00	80.56	0.286	0.624 (0.491 to 0.745) ^{NS}
"Any" SM + AT%	> 9.64	72.00	66.67	0.387	0.660 (0.527 to 0.776)
RWA SM Tonic%	> 0.83	40.00	83.33	0.233	0.578 (0.445 to 0.703) ^{NS}
RWA SM Phasic%	> 1.47	60.00	66.67	0.267	0.593 (0.460 to 0.717) ^{NS}
RWA SM%	> 3.21	64.00	72.22	0.362	0.644 (0.511 to 0.762) ^{NS}
RWA AT %	> 0.79	36.00	94.12	0.221	0.567 (0.434 to 0.694) ^{NS}
RWA SM + AT%	> 3.67	64.00	66.67	0.307	0.651 (0.518 to 0.768)
NT1 versus HC					
3-second mini-epoch					
Phasic SM%	> 7.69	60.00	94.12	0.541	0.818 (0.696 to 0.906)
"Any" SM%	> 8.05	68.00	91.18	0.592	0.811 (0.687 to 0.901)
Phasic AT%	> 2.03	84.00	91.18	0.752	0.920 (0.819 to 0.975)
"Any" SM + AT%	> 7.91	84.00	85.29	0.693	0.867 (0.753 to 0.941)
30-second epoch					
RWA SM Tonic%	> 0.78	40.00	88.24	0.282	0.638 (0.503 to 0.759)
RWA SM Phasic%	> 1.72	56.00	85.29	0.493	0.709 (0.577 to 0.820)
RWA SM%	> 3.11	64.00	85.29	0.478	0.746 (0.616 to 0.851)
RWA AT %	> 0.00	40.00	94.12	0.341	0.673 (0.538 to 0.789)
RWA SM + AT%	> 4.37	60.00	91.18	0.512	0.781 (0.654 to 0.878)
NT2 versus HC					
3-second mini-epoch					
Phasic SM%	> 2.88	91.67	50.00	0.417	0.739 (0.620 to 0.836)
"Any" SM%	> 3.19	88.89	50.00	0.389	0.730 (0.610 to 0.829)
Phasic AT%	> 1.74	75.00	88.24	0.632	0.859 (0.756 to 0.931)
"Any" SM + AT%	> 4.98	91.67	58.82	0.505	0.788 (0.673 to 0.876)
30-second epoch					
RWA SM Tonic%	> 0.00	33.33	85.29	0.186	0.585 (0.461 to 0.701) ^{NS}
RWA SM Phasic%	> 0.61	55.56	67.65	0.232	0.630 (0.506 to 0.742)
RWA SM%	> 2.16	44.44	79.41	0.239	0.628 (0.504 to 0.740) ^{NS}
RWA AT %	> 0.00	30.56	94.12	0.247	0.623 (0.499 to 0.736)
RWA SM + AT%	> 4.37	33.33	91.18	0.245	0.628 (0.523 to 0.760) ^{NS}

Note: Narcolepsy ($n = 61$), NT1 ($n = 25$), NT2 ($n = 36$), healthy controls ($n = 34$). All ROC curves had p-values < 0.05 unless indicated with ^{NS} (non-significant, $p > 0.05$)

NT Narcolepsy type, *HC* Healthy controls, *SM* Submentalis muscle, *AT* Anterior tibialis muscle, *REM* Rapid eye movement, *RWA* REM sleep without atonia, *AUC* Area under the curve, *CI* Confidence interval, *ROC* Receiver operating characteristic

activity during REM sleep, even when there are no clear symptoms of REM behavior disorder (RBD) [18–20]. In NT2, some studies suggest partial hypocretin pathway dysfunction [14, 15], evidenced by intermediate CSF orexin levels or neuronal loss in a subset of patients, which may contribute to milder REM sleep abnormalities compared to NT1. These findings align with prior studies [3, 4] and reinforce the importance of REM sleep motor dysregulation in understanding narcolepsy [21, 22].

PSG data confirmed key features of narcolepsy, including significantly shorter REM latency in NT2 compared to controls, as well as more frequent SOREMPs in both NT1 and NT2 [1, 23, 24]. Notably, while NT1 patients exhibited shorter REM latency compared to HC, this difference

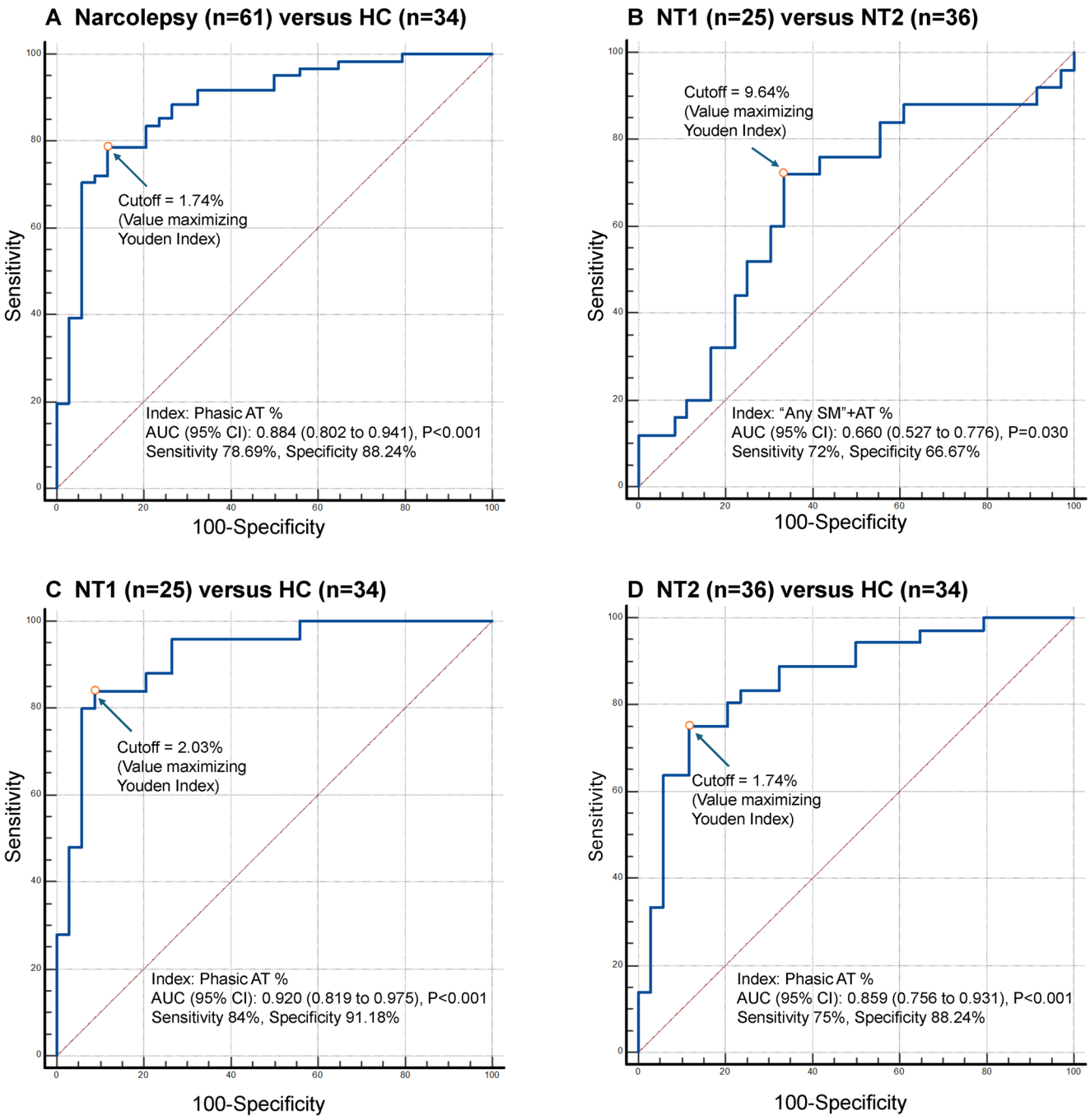

Fig. 3 A-D. Receiver operating characteristic (ROC) curves for the best-performing indices in four models. (**A**) Phasic AT % was the best-performing index in differentiating narcolepsy from HC. (**B**) The combined "any" SM+AT % index showed moderate performance in differentiating NT1 from NT2C. (**C**) Phasic AT % demonstrated excellent diagnostic accuracy in differentiating NT1 from HC. (**D**) Phasic AT % was again the best-performing index in differentiating NT2 from HC Cutoff values were determined by maximizing the Youden Index. AUC, sensitivity, and specificity values are provided for each ROC curve AUC: Area under the curve; CI: Confidence interval; NT: Narcolepsy type; HC: Healthy controls; SM: Submentalis muscle; AT: Anterior tibialis muscle

did not reach statistical significance, contrasting with previous reports [25], while PLMI was elevated in NT1 versus controls, aligning with prior findings [26]. This lack of significant difference in REML among NT1 and HC, despite its classic association with narcolepsy, may reflect the relatively small sample size of NT1 patients in our study. In our study, SOREMPs were observed in 32% of NT1 patients and 25% of NT2 patients. Although nocturnal SOREMPs are considered an important diagnostic criterion for narcolepsy [23, 24], their prevalence appears to be lower in

certain populations [27]. For example, studies have shown a reduced prevalence of SOREMPs both in nocturnal PSG and MSLT in the Chinese population [28, 29]. Additionally, factors such as shift work, sleep deprivation [17, 24], and other conditions that may lead to REM sleep rebound can alter the occurrence of SOREMPs [27, 30]. These variations limit the utility of SOREMPs as a diagnostic marker in such scenarios. In such scenarios, EMG-based measures of REM muscle activity may offer stable, objective biomarkers that are less influenced by external factors [3, 5, 6]. For instance, phasic AT activity identified 78.69% of narcolepsy patients (sensitivity) with a specificity of 88.24%, while combined SM+AT activity identified 70.49% of patients with a specificity of 82.35%. These EMG measures remained robust discriminators even when nocturnal SOREMPs were absent, underscoring their value in cases where SOREMP assessment is impractical or inconclusive [3, 5].

The superiority of the 3sRME method in our study lies in its ability to capture all transient phasic muscle activities which are more prevalent in narcolepsy than tonic activity [4, 19]. The 3sRME approach, by dividing 30sRE into 3sRME, provides higher temporal resolution and sensitivity to these short-duration events, which are often missed in 30sRE analyses when they do not fulfill the criteria for positive RWA threshold (e.g., ≥5 phasic bursts per 30 s epoch) [8]. For example, phasic AT activity in 3sRME achieved the highest diagnostic accuracy (AUC = 0.884 for narcolepsy vs. HC), as it detects all subtle lower-limb movements that RWA quantified using 30sRE thresholds may overlook. This detailed view is crucial for telling apart different narcolepsy types and highlights the limitations of 30sRE methods in capturing all brief muscle events, which are more common in narcolepsy [4, 19].

In contrast, RWA indices derived from 30sRE showed limited discriminative power in narcolepsy. In group comparison and logistic regression, RWA scored with phasic SM, total SM, and combined SM+AT were significantly elevated in NT1 versus HC, they failed to differentiate NT2 from HC or NT1 from NT2, consistent with prior research [3, 22]. Despite the lack of significance in group comparisons and logistic regression, ROC curve analysis revealed that RWA due to phasic SM had significant AUC values for differentiating NT2 from HC. Additionally, tonic SM activity did not differ between groups and did not yield a meaningful cutoff for narcolepsy and NT2, which further emphasizes the importance of phasic activity in narcolepsy [19]. While the RWA index for combined SM+AT (AUC = 0.691) provided marginal utility under AASM criteria for narcolepsy, their performance remained inferior to 3sRME measures. These results advocate for integrating REM sleep phasic muscle activity, both SM and AT into diagnostic frameworks for quantification in narcolepsy. However, these results should be interpreted with caution due to their potential confounding effects. While the inclusion of phasic AT activity improved the differentiation of narcolepsy in our study, previous research studies suggest that the AT is less specific for RWA quantification, as it may contain additional EMG activity such as periodic limb movements (PLMs) and fragmentary myoclonus [11, 12]. Overall, 3sRME quantification refines narcolepsy diagnosis by capturing dynamic REM sleep disturbances that RWA quantification using 30sRE may miss.

Consistent with Khalil et al. [6], our study identified phasic REM activity as a more reliable marker for narcolepsy compared to controls. This aligns with Manni et al.'s interpretation of such activity as a motor phenomenon during REM sleep, analogous to dream enactment in RBD [31]. However, our reported tonic RWA percentages were notably lower than those in previous studies. For example, Bin-Hasan et al. (2018) observed a median tonic RWA of 6.4% in pediatric NT1 patients, while Yon et al. (2021) reported tonic RWA indices exceeding 5% in adult NT1 patients [3, 32]. We attribute this discrepancy to methodological differences: these studies scored epochs as a tonic if chin EMG amplitude exceeded the minimum NREM amplitude for ≥50% of the epoch, whereas our study required amplitudes twice the background activity.

Recent studies have explored the use of automatic algorithms for quantifying muscle activity during REM sleep in narcolepsy, offering promising avenues for improving the efficiency and scalability of RWA detection [33, 34]. Automatic algorithms have reported strong discrimination between NT1 and controls and correlations with CSF hypocretin levels [33]. These methods aim to overcome the limitations of manual scoring, which is time-consuming and requires specialized training [25]. However, most published algorithms remain in the research domain, and there are no consensus guidelines [11]. While our study employed manually scored 3 s and 30 s EMG metrics, our results are broadly consistent with these automated findings, highlighting elevated phasic and overall EMG activity in NT1 and subtler changes in NT2. We believe that both manual and automated EMG scoring should advance in parallel that manual scoring continues to serve as a robust reference framework, while further development and validation of automated tools are essential for enabling broader clinical integration and real-world applicability. Automated approaches may be particularly useful in high-throughput or resource-limited settings. As this field evolves, future studies should aim to harmonize manual and automatic methodologies to improve both reliability and accessibility of EMG-based narcolepsy assessment.

The findings of this study highlight the potential of EMG activity indices as alternative biomarkers for narcolepsy

[3, 6]. While traditional diagnostic tools such as the MSLT and cerebrospinal fluid (CSF) hypocretin measurements as well as SOREMPs remain valuable, they are often limited by accessibility, cost, and variability [3, 5, 35]. The MSLT, for example, requires an additional test on the following day, which can be burdensome for patients and challenging to coordinate. Similarly, CSF hypocretin testing is invasive and not widely available, particularly in resource-limited settings. SOREMPs, while useful, may not always be detectable in all patients, limiting their diagnostic utility in certain cases. By integrating findings of EMG activity into clinical practice, clinicians can leverage routine PSG data to improve the identification and differentiation of narcolepsy subtypes.

Clinical implication

Elevated EMG activity and RWA in narcolepsy patients identified through 3sRME and AASM-positive RWA in 30sRE methods have clinical relevance. This approach is especially useful when patients present with clinical symptoms suggestive of narcolepsy, such as EDS, cataplexy, sleep paralysis, hypnagogic/hypnopompic hallucinations, fragmented nighttime sleep, and automatic behaviors. These symptoms, combined with EMG activity findings, can help clinicians identify and differentiate narcolepsy subtypes more effectively when CSF hypocretin testing or MSLT is inaccessible [3, 35]. The 3sRME approach was found to be essential for differentiating narcolepsy types from controls as the 30sRE approach may overlook some EMG activity. In clinical practice, distinguishing NT2 from other conditions associated with excessive daytime sleepiness can be challenging, particularly in the absence of definitive biomarkers such as CSF hypocretin levels or SOREMPs on MSLT [36]. Both Yon et al. and Bin-Hasan et al. noted that RWA is significantly higher in NT1 but often overlaps between NT2 and controls [3, 32]. The present study supports this pattern and found that the 3sRME method can be used for diagnostic discrimination of NT2 in real-world settings. The EMG activity during REM sleep can also provide an alternative marker when assessment with SOREMPs is not viable. By integrating EMG activity indices into routine PSG evaluations, clinicians can enhance screening accuracy and improve patient care, particularly when traditional diagnostic methods are unavailable or inconclusive.

Limitations and future directions

This study has limitations, including a small sample size which may affect generalizability. Another limitation of our study is its retrospective design, which may introduce biases related to confounding factors such as co-morbid sleep disorders or medication use. Future prospective studies should address these gaps to validate our findings.

Conclusion

The study revealed elevated levels of REM sleep EMG activity in narcolepsy patients compared to HC, with more pronounced abnormalities observed in NT1 than in NT2. The 3sRME quantification approach proved more effective in differentiating narcolepsy and its subtypes from HC compared to the AASM-defined positive RWA epoch method. These results suggest that EMG activity indices during REM sleep may serve as potential alternative biomarkers for narcolepsy, highlighting the importance of methodological precision in assessing REM sleep abnormalities.

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Data availability The data from the CCSD is accessible via psg.wanglitech.com. Researchers interested in accessing the data must contact the Department of Psychiatry (Sleep Medicine Center) at Nanfang Hospital and submit a detailed research proposal outlining a clear scientific objective. Following an internal review process, approved researchers will be granted an account and password to access the database.

Declarations

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the ethical committee of Nanfang Hospital, Southern Medical University and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Disclosure statement Financial Disclosure: none. Non-financial Disclosure: none.

Informed consent Since this study is retrospective in nature and involves the analysis of existing anonymized data, for this type of study formal consent is not required.

Conflict of interest The authors declare that they have no conflict of interest.

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